

MySage Reusable Module Demo Library

Demo Module Selected

COM-001 — Email Sending Module

This module is selected because it is:

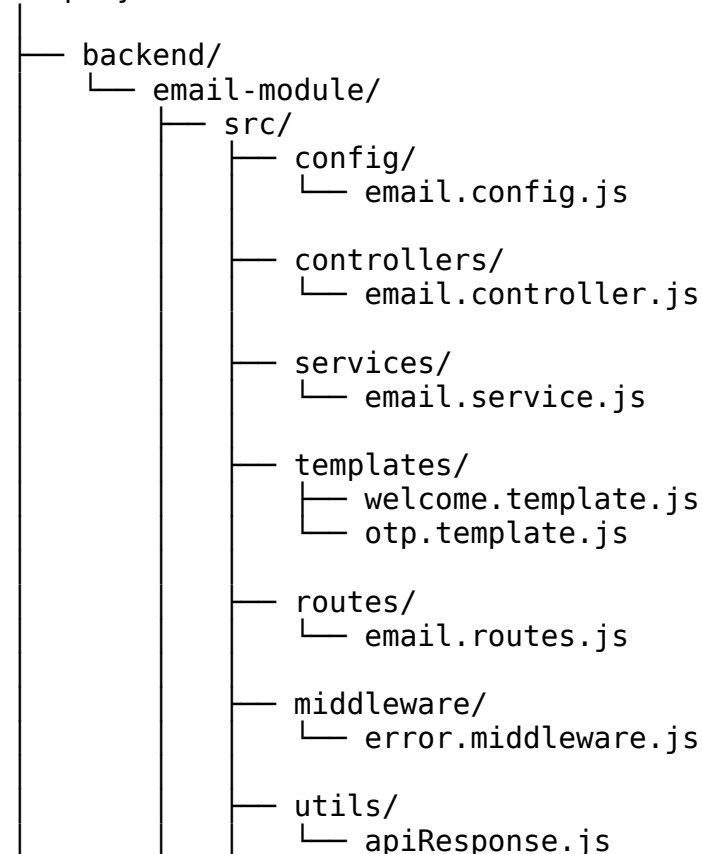
- High reusability across all products
- Easy to demonstrate independently
- Commonly required in all enterprise applications
- API-ready and scalable
- Suitable for GitHub showcase and reusable architecture demonstration

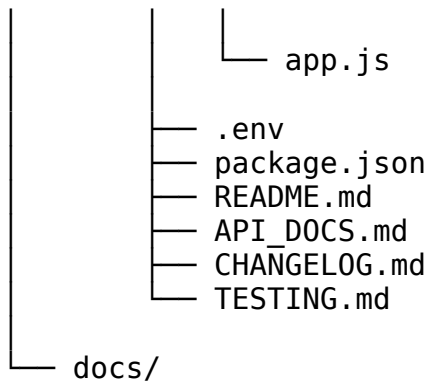
Based on the reusable module strategy documents provided.

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Recommended GitHub Repository Structure

company-shared-modules/





Technology Stack

Layer	Technology
Backend	Node.js
API	Express.js
Email Engine	Nodemailer
Template Engine	HTML Templates
Authentication	SMTP
Documentation	Swagger/OpenAPI
Testing	Jest

package.json

```
{
  "name": "@mysage/email-module",
  "version": "1.0.0",
  "description": "Reusable enterprise email module",
  "main": "src/app.js",
  "scripts": {
    "start": "node src/app.js",
    "dev": "nodemon src/app.js",
    "test": "jest"
  },
  "dependencies": {
    "cors": "^2.8.5",
    "dotenv": "^16.4.5",
    "express": "^4.19.2",
    "nodemailer": "^6.9.14"
  },
  "devDependencies": {
    "jest": "^29.7.0",
  }
}
```

```
    "nodemon": "^3.1.4"
  }
}
```

.env

```
PORT=5000
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your-email@gmail.com
SMTP_PASS=your-app-password
FROM_EMAIL=no-reply@mysage.com
```

src/config/email.config.js

```
const nodemailer = require('nodemailer');
require('dotenv').config();

const transporter = nodemailer.createTransport({
  host: process.env.SMTP_HOST,
  port: process.env.SMTP_PORT,
  secure: false,
  auth: {
    user: process.env.SMTP_USER,
    pass: process.env.SMTP_PASS,
  },
});

module.exports = transporter;
```

src/utlis/apiResponse.js

```
const successResponse = (res, message, data = {}) => {
  return res.status(200).json({
    success: true,
    message,
    data,
  });
};

const errorResponse = (res, message, error = {}) => {
  return res.status(500).json({
    success: false,
```

```
        message,  
        error,  
    });  
};  
  
module.exports = {  
    successResponse,  
    errorResponse,  
};
```

src/templates/welcome.template.js

```
const welcomeTemplate = (name) => {  
    return `  
        <div style="font-family: Arial; padding:20px;">  
            <h2>Welcome to MySage</h2>  
            <p>Hello ${name},</p>  
            <p>Your account has been successfully created.</p>  
            <p>Thank you for joining MySage Platform.</p>  
        </div>  
    `;  
};  
  
module.exports = welcomeTemplate;
```

src/templates/otp.template.js

```
const otpTemplate = (otp) => {  
    return `  
        <div style="font-family: Arial; padding:20px;">  
            <h2>Email Verification</h2>  
            <p>Your OTP is:</p>  
            <h1>${otp}</h1>  
            <p>This OTP is valid for 10 minutes.</p>  
        </div>  
    `;  
};  
  
module.exports = otpTemplate;
```

src/services/email.service.js

```
const transporter = require('../config/email.config');
const welcomeTemplate = require('../templates/welcome.template');
const otpTemplate = require('../templates/otp.template');

const sendWelcomeEmail = async (email, name) => {
  const mailOptions = {
    from: process.env.FROM_EMAIL,
    to: email,
    subject: 'Welcome to MySage',
    html: welcomeTemplate(name),
  };

  return await transporter.sendMail(mailOptions);
};

const sendOTPEmail = async (email, otp) => {
  const mailOptions = {
    from: process.env.FROM_EMAIL,
    to: email,
    subject: 'OTP Verification',
    html: otpTemplate(otp),
  };

  return await transporter.sendMail(mailOptions);
};

module.exports = {
  sendWelcomeEmail,
  sendOTPEmail,
};
```

src/controllers/email.controller.js

```
const {
  sendWelcomeEmail,
  sendOTPEmail,
} = require('../services/email.service');

const {
  successResponse,
  errorResponse,
} = require('../utils/apiResponse');

const sendWelcome = async (req, res) => {
  try {
```

```
    const { email, name } = req.body;

    await sendWelcomeEmail(email, name);

    return successResponse(res, 'Welcome email sent successfully');
  } catch (error) {
    return ErrorResponse(res, 'Failed to send welcome email', error.message);
  }
};

const sendOTP = async (req, res) => {
  try {
    const { email, otp } = req.body;

    await sendOTPEmail(email, otp);

    return successResponse(res, 'OTP email sent successfully');
  } catch (error) {
    return ErrorResponse(res, 'Failed to send OTP email', error.message);
  }
};

module.exports = {
  sendWelcome,
  sendOTP,
};
```

src/routes/email.routes.js

```
const express = require('express');
const router = express.Router();

const {
  sendWelcome,
  sendOTP,
} = require('../controllers/email.controller');

router.post('/welcome', sendWelcome);
router.post('/otp', sendOTP);

module.exports = router;
```

src/middleware/error.middleware.js

```
const errorHandler = (err, req, res, next) => {  
  console.error(err.stack);  
  
  res.status(500).json({  
    success: false,  
    message: 'Internal Server Error',  
  });  
};  
  
module.exports = errorHandler;
```

src/app.js

```
const express = require('express');  
const cors = require('cors');  
require('dotenv').config();  
  
const emailRoutes = require('./routes/email.routes');  
const errorHandler = require('./middleware/error.middleware');  
  
const app = express();  
  
app.use(cors());  
app.use(express.json());  
  
app.use('/api/email', emailRoutes);  
  
app.use(errorHandler);  
  
const PORT = process.env.PORT || 5000;  
  
app.listen(PORT, () => {  
  console.log(`Email Module Running on Port ${PORT}`);  
});
```

API Endpoints

Send Welcome Email

POST /api/email/welcome

Request

```
{
  "email": "user@example.com",
  "name": "Vineet"
}
```

Response

```
{
  "success": true,
  "message": "Welcome email sent successfully"
}
```

Send OTP Email

POST /api/email/otp

Request

```
{
  "email": "user@example.com",
  "otp": "456789"
}
```

Response

```
{
  "success": true,
  "message": "OTP email sent successfully"
}
```

README.md

MySage Reusable Email Module

Reusable enterprise email module built using Node.js and Express.

Features

- Welcome Emails
- OTP Emails
- SMTP Support
- HTML Templates
- Standard API Responses
- Modular Architecture
- Easy Integration

Installation

npm install

Run Application

npm run dev

API Base URL

http://localhost:5000/api/email

CHANGELOG.md

Changelog

v1.0.0

- Initial reusable email module release
 - SMTP support
 - Welcome email template
 - OTP template
 - REST APIs
-

TESTING.md

Testing Guide

Run Tests

npm test

Suggested Test Cases

- Send welcome email
 - Send OTP email
 - Invalid email validation
 - SMTP failure handling
 - Empty payload validation
-

Frontend React Integration Example

Recommended Frontend Structure

```
frontend/
├── src/
│   ├── api/
│   │   └── emailApi.js
│   ├── components/
│   │   ├── SendWelcomeEmail.jsx
│   │   └── SendOTPEmail.jsx
│   ├── App.jsx
│   └── main.jsx
├── package.json
└── vite.config.js
```

Frontend package.json

```
{
  "name": "mysage-email-demo-ui",
  "private": true,
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build"
  },
  "dependencies": {
    "axios": "^1.7.2",
    "react": "^18.3.1",
    "react-dom": "^18.3.1"
  },
  "devDependencies": {
    "@vitejs/plugin-react": "^4.3.1",
    "vite": "^5.3.4"
  }
}
```

src/api/emailApi.js

```
import axios from 'axios';

const API = axios.create({
  baseURL: 'http://localhost:5000/api/email',
});

export const sendWelcomeEmail = async (payload) => {
  return await API.post('/welcome', payload);
};

export const sendOTPEmail = async (payload) => {
  return await API.post('/otp', payload);
};
```

src/components/SendWelcomeEmail.jsx

```
import React, { useState } from 'react';
import { sendWelcomeEmail } from '../api/emailApi';

const SendWelcomeEmail = () => {
  const [formData, setFormData] = useState({
    email: '',
    name: '',
  });

  const [message, setMessage] = useState('');

  const handleChange = (e) => {
    setFormData({
      ...formData,
      [e.target.name]: e.target.value,
    });
  };

  const handleSubmit = async (e) => {
    e.preventDefault();

    try {
      const response = await sendWelcomeEmail(formData);
      setMessage(response.data.message);
    } catch (error) {
      setMessage('Failed to send welcome email');
    }
  };
};
```

```

    return (
      <div style={{ padding: '20px', border: '1px solid #ddd',
marginBottom: '20px' }}>
        <h2>Send Welcome Email</h2>

        <form onSubmit={handleSubmit}>
          <input
            type="text"
            name="name"
            placeholder="Enter Name"
            value={formData.name}
            onChange={handleChange}
          />

          <br /><br />

          <input
            type="email"
            name="email"
            placeholder="Enter Email"
            value={formData.email}
            onChange={handleChange}
          />

          <br /><br />

          <button type="submit">Send Welcome Email</button>
        </form>

        <p>{message}</p>
      </div>
    );
  };
};

export default SendWelcomeEmail;

```

src/components/SendOTPEmail.jsx

```

import React, { useState } from 'react';
import { sendOTPEmail } from '../api/emailApi';

const SendOTPEmail = () => {
  const [formData, setFormData] = useState({
    email: '',
    otp: '',
  });

```

```

const [message, setMessage] = useState('');

const handleChange = (e) => {
  setFormData({
    ...formData,
    [e.target.name]: e.target.value,
  });
};

const handleSubmit = async (e) => {
  e.preventDefault();

  try {
    const response = await sendOTPEmail(formData);
    setMessage(response.data.message);
  } catch (error) {
    setMessage('Failed to send OTP email');
  }
};

return (
  <div style={{ padding: '20px', border: '1px solid #ddd' }}>
    <h2>Send OTP Email</h2>

    <form onSubmit={handleSubmit}>
      <input
        type="email"
        name="email"
        placeholder="Enter Email"
        value={formData.email}
        onChange={handleChange}
      />

      <br /><br />

      <input
        type="text"
        name="otp"
        placeholder="Enter OTP"
        value={formData.otp}
        onChange={handleChange}
      />

      <br /><br />

      <button type="submit">Send OTP Email</button>
    </form>

    <p>{message}</p>
  </div>
);

```

```
        </div>
    );
};

export default SendOTPEmail;
```

src/App.jsx

```
import React from 'react';
import SendWelcomeEmail from './components/SendWelcomeEmail';
import SendOTPEmail from './components/SendOTPEmail';

const App = () => {
    return (
        <div style={{ maxWidth: '600px', margin: '50px auto' }}>
            <h1>MySage Email Module Demo</h1>

            <SendWelcomeEmail />
            <SendOTPEmail />
        </div>
    );
};

export default App;
```

src/main.jsx

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

ReactDOM.createRoot(document.getElementById('root')).render(
    <React.StrictMode>
        <App />
    </React.StrictMode>
);
```

Running Backend

```
cd backend/email-module
npm install
npm run dev
```

Running Frontend

```
cd frontend
npm install
npm run dev
```

Sample Demo Flow

1. User enters email and name
 2. React frontend calls backend API
 3. Node.js email module processes request
 4. SMTP service sends email
 5. API response returned to frontend
 6. Success message displayed
-

Architecture Flow

```
React Frontend
  ↓
Axios API Layer
  ↓
Express Routes
  ↓
Controller Layer
  ↓
Reusable Email Service
  ↓
SMTP Provider
  ↓
Email Delivered
```

GitHub Upload Recommendation

Repository Name

mysage-reusable-email-module

Recommended Tags

nodejs
react
reusable-library
enterprise-module
email-module
shared-library
expressjs
smtp
nodemailer
mysage

Suggested README Screenshot Sections

Add screenshots for:

- Welcome Email UI
 - OTP Email UI
 - Postman Testing
 - Swagger API
 - GitHub Folder Structure
-

Enterprise Improvements Recommended

Security

- Rate limiting
- Input validation
- JWT security
- Environment encryption
- SMTP credential vaulting

Scalability

- Queue processing
- Worker threads
- Redis caching
- Email batching
- Retry queues

DevOps

- Docker support

- GitHub Actions
 - Kubernetes deployment
 - SonarQube integration
 - CI/CD pipelines
-

Suggested Future Enhancements

Phase 2

- Queue-based email processing
- RabbitMQ integration
- Retry mechanisms
- Bulk email sending
- Dynamic template builder
- Email analytics
- Delivery tracking

Phase 3

- AI-generated email templates
 - Drag-and-drop email designer
 - Multi-language support
 - Multi-tenant email branding
 - Campaign management
-

GitHub README Banner Content

MySage Shared Modules Library

Enterprise-grade reusable modules architecture for scalable applications.

Available Modules

- Authentication Module
- RBAC Module
- Email Module
- Notification Module
- Logging Module
- Workflow Engine
- Dashboard Components
- Reporting Engine

Tech Stack

- Node.js
- React
- Supabase
- GitHub Actions
- Docker
- Tailwind

Goal

Accelerate product delivery through reusable enterprise architecture.

Why This Module Is Ideal For Demo

This module aligns strongly with the reusable architecture strategy because:

- It demonstrates modular backend architecture
 - Shows reusable service patterns
 - Demonstrates standardized API responses
 - Easy to integrate into multiple products
 - Simple to deploy independently
 - Can later become npm package
 - Can later integrate with workflow engine
 - Easy to demonstrate in GitHub portfolio
-

Recommended Next Modules After This

1. Authentication Module
 2. RBAC Module
 3. Notification Module
 4. File Upload Module
 5. Dashboard Widget Module
 6. Workflow Engine
 7. Rule Engine
 8. Audit Log Module
-

Alignment With Strategy Documents

This implementation follows:

- High Reusability classification
- Standard repository structure
- Standard documentation structure
- Semantic versioning strategy
- API-ready modular architecture
- Shared reusable library concept
- Modular ownership strategy

Referenced from uploaded strategy documents.   
  